

CS 101: Computer Science & Python Programming

Module 1: Introduction to Computer Science and Python Syntax

Computer science is the study of computation, information processing, and algorithmic problem-solving. Python is a high-level, interpreted programming language widely used in software engineering, data analysis, and artificial intelligence.

Variables in Python serve as named references to objects stored in dynamic memory. Unlike statically typed languages like Java or C++, Python uses dynamic typing, meaning a variable's type is determined at runtime based on the assigned object.

Primitive data types in Python include integers (int), floating-point numbers (float), booleans (bool), and strings (str). Data structures such as lists, tuples, sets, and dictionaries allow developers to organize complex data.

Module 2: Control Flow Structures and Functions

Control flow structures dictate the order in which code statements execute. Conditional statements (if, elif, else) evaluate boolean expressions to branch program execution.

Iteration is handled via for-loops and while-loops. A for-loop iterates over sequences such as lists or ranges, whereas a while-loop continues executing as long as a specified condition remains true.

Functions modularize code into reusable blocks. Defined using the 'def' keyword, functions can accept positional and keyword arguments and return values using the 'return' statement. Scope rules in Python follow the LEGB rule (Local, Enclosing, Global, Built-in).

Module 3: Object-Oriented Programming (OOP) Principles

Object-Oriented Programming (OOP) is a design paradigm built around objects containing data and behavior. The core pillars of OOP are Encapsulation, Abstraction, Inheritance, and Polymorphism.

A Class acts as a blueprint for creating objects. The '`__init__`' initializer method initializes an instance's attributes upon creation. Encapsulation hides internal object implementation details, exposing only necessary interfaces through public methods.